Sai Krishna Pothana

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PROFESSIONAL SUMMARY

Data Engineer with 4+ years of experience building production-grade pipelines processing 5TB+ daily and 100K+ events per minute, with two live deployed data and AI projects. Core stack: Python, SQL, Spark/PySpark, Kafka, Airflow, Snowflake, Databricks, and dbt across AWS, Azure, and GCP. Delivered 40% latency reduction, 50% fewer data errors, and 35% faster queries through data quality engineering, dimensional modeling, and CI/CD automation. Leads architectural design discussions, mentors engineers, and builds AI-powered data tools.

TECHNICAL SKILLS

Languages & Libraries	Python, SQL, R, PySpark, Bash/Shell Scripting, Pandas, NumPy, Matplotlib
Core Data Engineering	Apache Spark, Spark Structured Streaming, Apache Kafka, Apache Airflow, ETL, ELT, SSIS, Data Pipelines, Real-Time Streaming, Batch Processing, REST API, Dimensional Modeling
Modern Data Platform	Snowflake, Databricks, dbt, dbt Cloud, Microsoft Fabric (OneLake, Delta Parquet), Matillion, Delta Lake, Apache Iceberg, Data Warehousing, Data Lakehouse
Cloud Platforms	AWS (S3, Glue, Redshift, EMR, MSK, Lambda, EKS, CloudWatch, Lake Formation), Azure (Data Factory, ADLS, Synapse Analytics, Databricks, Event Hubs), GCP (BigQuery, Dataflow, Pub/Sub)
Data Quality & Governance	Great Expectations, Data Quality, Data Governance, Data Lineage, Unity Catalog, DataOps, Pipeline Monitoring, SLA Management
Databases	PostgreSQL, MySQL, SQL Server, MongoDB, DynamoDB, Spark SQL
BI & Visualization	Power BI, Tableau, KNIME, Matplotlib, Pandas
DevOps & Infrastructure	Docker, Kubernetes, Terraform, GitHub Actions, Git, CI/CD Pipelines
AI & ML	Generative AI, LLMs, Retrieval Augmented Generation (RAG), Machine Learning, Scikit-learn, NLP

PROFESSIONAL EXPERIENCE

Data Engineer | Datics Inc | Cincinnati, OH May 2025 – Jun 2026

- Designed ETL/ELT pipelines on AWS using Python, SQL, Glue, and Lambda, ingesting data from 10+ source systems into a scalable S3 data lake, reducing end-to-end latency by 40%.
- Built Snowflake data warehouse schemas using star and snowflake dimensional modeling with clustering keys, improving query performance by 35%; developed dbt Cloud and Matillion ELT models with automated testing and documentation.
- Reduced downstream data errors by 50% by implementing a Great Expectations validation layer in CI/CD, after tracing a recurring null and malformed value spike breaking downstream joins.
- Engineered Apache Kafka producer/consumer pipelines on AWS MSK for real-time ingestion, scaling to 100K+ events/minute with fault tolerance via consumer group design and dead-letter queues.
- Built large-scale PySpark and Spark Structured Streaming workflows on AWS EMR and Databricks, processing 5TB+ daily using partitioning, caching, and broadcast joins.
- Built and maintained SSIS packages for enterprise ETL operations, enforcing data governance and access policies via Unity Catalog on Databricks.
- Orchestrated Apache Airflow DAGs with automatic retries, dependency management, and CloudWatch SLA alerting, maintaining 99%+ pipeline reliability in production.
- Led Snowflake architecture design discussions and mentored 2 junior data engineers on dbt, Airflow, and data quality best practices.

Data Engineer | Accenture | Hyderabad, India Jun 2023 – Jul 2024

- Implemented Microsoft Fabric pipelines storing data in OneLake in Delta Parquet format, orchestrating workflows across Azure Data Factory and Spark Notebooks, and reducing report delivery time by 30% for downstream Power BI consumers.
- Maintained ETL pipelines using Azure Data Factory, Matillion, and Azure Data Lake Storage, validating field mappings and resolving data integrity issues across financial and operational reporting.
- Performed data extraction, validation, and transformation using SQL and Python across Azure SQL, MySQL, and MongoDB for cross-functional stakeholder reporting.
- Optimized SQL query performance and indexing in PostgreSQL and MySQL, cutting data retrieval times by 30% and improving dashboard load performance.
- Built operational dashboards using Power BI, R, and Matplotlib, surfacing KPI insights on project margins, service utilization, and customer metrics for product and operations teams.

- Collaborated with finance, QA, and product teams in Agile sprints to gather requirements and deliver analytics-ready datasets.

Junior Data Engineer | **ASICS Technologies | Hyderabad, India** May 2021 – May 2023

- Developed PySpark jobs for distributed processing on Hadoop and Azure Databricks, improving pipeline throughput by 25% via predicate pushdown and optimized partitioning.
- Owned Python and SQL-based ETL development for internal reporting, integrating data from multiple business domains into a centralized data warehouse for daily BI reporting.
- Deployed data services using Docker on AWS EC2/S3 with GitHub Actions CI/CD, reducing deployment time by 20%; managed container orchestration with Kubernetes and infrastructure provisioning with Terraform.
- Implemented Apache Airflow DAGs for pipeline orchestration, automating dependency management, failure recovery, and SLA alerting across critical workflows.
- Diagnosed and resolved Kafka consumer lag and partition rebalancing on production pipelines, restoring throughput within SLA windows.

PROJECTS

AI-Powered BigQuery Data Catalog Assistant | bigquery-rag.streamlit.app

- Built and deployed an AI-powered data catalog assistant using RAG (TF-IDF, cosine similarity) and Llama 3.3 70B via Groq, enabling natural language table discovery, schema explanation, and SQL generation over a 7-table, 68-column BigQuery dataset.
- Implemented cost and safety guardrails for LLM-generated queries: read-only access, automatic LIMIT 100, and 200MB byte cap; deployed as a public web app on Streamlit Cloud.

Tech Stack: Python, RAG, Scikit-learn, LLaMA 3.3 70B (Groq), Google BigQuery, SQL, Streamlit

Real-Time Stock Market Data Pipeline | stock-rag-pipeline.vercel.app

- Architected a real-time streaming pipeline ingesting live stock market data via Apache Kafka producers and consumers, processing events with Spark Structured Streaming.
- Persisted enriched data to AWS S3 and PostgreSQL for structured querying; built Tableau dashboards for live market trend visualization with sub-second latency.

Tech Stack: Apache Kafka, Spark Structured Streaming, AWS S3, PostgreSQL, Tableau, Python

EDUCATION

Master of Engineering in Computer Engineering | University of Cincinnati | Cincinnati, OH | Aug 2024 – Apr 2026

Relevant Coursework: Big Data Analytics, Cloud Computing, Distributed Systems, Machine Learning, Data Warehousing

CERTIFICATIONS & PUBLICATION

- Databricks Certified Data Engineer Professional
- Data Engineering Essentials – IBM (Coursera)
- Data Analysis with Python – IBM (Cognitive Class)
- AI Fluency: Framework & Foundations – Anthropic

Publication:

Pothana, S.K., et al. (2024). Analysis of Satellite Images for Disaster Detection Using CNN. International Journal of Applied Science Engineering and Management (IJASEM), Vol. 18, Issue 1, pp. 267-270. ISSN: 2454-9940. www.ijasem.org